



Test 1 FSPM 0024 Calculus 2019 2020-3

CALCULUS - foundation (Universiti Teknologi Malaysia)



messages.pdf_cover_qr_code_label

Foundation Programme
Universiti Teknologi Malaysia
Session 2019/2020 Semester 3

Test 1 (15%)

Time: 75 minutes

FSPM 0024 Calculus

Answer all questions.

1. Evaluate the limit for each of the following.

(a) $\lim_{x \rightarrow 0} \frac{\sqrt{x+2} - \sqrt{2}}{x}$. (3 marks)

(b) $\lim_{x \rightarrow \infty} \frac{5x^2 + 2x + 1}{\sqrt{x^4 + 2}}$. (3 marks)

2. Use First Principle Formula to find $f'(x)$ if $f(x) = 2\sqrt{x+10}$. (4 marks)

3. Use appropriate rules to find $\frac{dy}{dx}$ for each of the following.

(a) $y = \operatorname{cosec}(\ln x)$. (3 marks)

(b) $y = \frac{\tan 2x}{e^{x^4}}$. (4 marks)

4. Given $\cos(xy) = 3y^2 - 2e^x$. Using implicit differentiation, find $\frac{dy}{dx}$ at point $(0, 1)$. (4 marks)

5. Given a set of parametric equations

$$x = 2t^2 + 1, \quad \text{and} \quad y = 2\sin(t^2).$$

Find $\frac{dy}{dx}$. Hence, show that $\frac{d^2y}{dx^2} = -\frac{1}{2}\sin(t^2)$. (5 marks)

6. Use differential to find the approximate value of

$$(16.1)^{\frac{1}{4}} + (16.1)^{\frac{1}{2}},$$

accurate to four decimal places.

(4 marks)

LIST OF FORMULA

Trigonometric
$\cos^2 x + \sin^2 x = 1$ $1 + \tan^2 x = \sec^2 x$ $\cot^2 x + 1 = \operatorname{cosec}^2 x$ $\sin 2x = 2 \sin x \cos x$ $\cos 2x = \cos^2 x - \sin^2 x$ $= 2 \cos^2 x - 1$ $= 1 - 2 \sin^2 x$ $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$ $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$ $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$ $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$ $2 \sin x \cos y = \sin(x + y) + \sin(x - y)$ $2 \sin x \sin y = -\cos(x + y) + \cos(x - y)$ $2 \cos x \cos y = \cos(x + y) + \cos(x - y)$
Logarithm
$a^x = e^{x \ln a}$ $\log_a x = \frac{\log_b x}{\log_b a}$

Differentiations
$\frac{d}{dx}[k] = 0, \quad k \text{ constant}$
$\frac{d}{dx}[x^n] = nx^{n-1}$
$\frac{d}{dx}[e^x] = e^x$
$\frac{d}{dx}[\ln x] = \frac{1}{x}$
$\frac{d}{dx}[\cos x] = -\sin x$
$\frac{d}{dx}[\sin x] = \cos x$
$\frac{d}{dx}[\tan x] = \sec^2 x$
$\frac{d}{dx}[\cot x] = -\operatorname{cosec}^2 x$
$\frac{d}{dx}[\sec x] = \sec x \tan x$
$\frac{d}{dx}[\operatorname{cosec} x] = -\operatorname{cosec} x \cot x$